

ELEFSIS, Greece

WMO: 167180

Lat: 38.064N

Lon: 23.556E

Elev: 44

StdP: 100.80

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.2	-6.9	2.1	3.9	-4.2	2.7	5.1	12.6	7.2	10.4	7.7	3.4	0	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	37.0	20.7	35.3	20.5	34.1	20.2	22.7	31.5	22.1	31.3	21.5	30.9	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	26.5	19.1	13.9	26.1	18.1	13.1	26.0	67.2	31.3	64.8	31.0	62.6	30.8	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.7	8.5	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-1.6	40.5	1.8	2.2	-2.9	42.1	-4.0	43.4	-5.0	44.7	-6.3	46.3
			-3.4	24.0	1.9	0.7	-4.7	24.6	-5.8	25.0	-6.8	25.4	-8.2	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	9.4	9.9	12.1	15.7	20.9	25.9	28.8	28.8	24.3	19.4	14.8	10.8
	DBStd	7.56	3.14	3.16	3.01	2.79	2.85	2.91	2.10	2.17	2.75	3.05	3.03	3.24
	HDD10.0	133	48	37	14	1	0	0	0	0	0	0	3	30
	HDD18.3	1171	277	237	194	86	8	0	0	0	1	24	110	234
	CDD10.0	3217	29	33	79	172	339	477	581	582	430	293	148	54
	CDD18.3	1213	0	0	1	7	89	227	323	324	181	58	5	0
	CDH23.3	13064	0	0	4	61	689	2436	4088	4009	1497	268	13	0
	CDH26.7	6051	0	0	0	9	187	1109	2121	2038	546	41	1	0
Wind	WSAvg	3.1	3.0	3.0	2.9	2.8	3.0	3.3	3.7	3.9	3.1	3.0	2.6	2.8
Precipitation	PrecAvg	425	58	47	45	25	19	10	7	7	23	43	65	77
	PrecMax	658	126	139	147	62	57	36	33	28	119	134	153	184
	PrecMin	218	4	11	2	1	1	0	0	0	0	1	5	11
	PrecStd	108	36	32	35	17	15	11	9	8	29	33	41	45
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	20.0	23.2	28.0	33.0	38.0	39.8	38.9	35.1	30.0	25.0	20.9
		MCWB	13.2	13.2	13.7	15.4	18.6	19.8	20.9	20.7	19.7	18.6	17.2	15.6
	2%	DB	17.1	18.1	21.0	25.0	30.2	35.2	37.0	37.0	33.0	27.8	23.0	19.0
		MCWB	12.6	12.4	13.4	15.2	17.6	20.2	20.7	20.8	19.8	18.3	17.0	14.5
	5%	DB	16.0	16.9	19.1	23.1	28.8	33.9	35.6	35.2	31.2	26.1	21.2	17.8
		MCWB	12.2	12.2	12.6	14.4	17.2	19.8	20.6	20.6	19.5	18.0	16.2	13.7
	10%	DB	14.9	15.2	17.9	21.8	27.0	32.1	34.1	34.0	29.9	24.8	20.0	16.1
		MCWB	11.1	11.2	12.1	13.9	16.7	19.5	20.2	20.3	19.1	17.7	15.5	12.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	14.5	15.2	17.4	19.8	22.2	23.5	23.9	22.7	21.4	19.4	16.8
		MCDB	17.6	18.3	20.6	24.3	28.6	32.2	31.5	31.6	29.2	25.5	22.9	19.6
	2%	WB	13.4	13.4	14.2	16.1	19.0	21.5	22.6	22.8	21.7	20.4	17.9	15.3
		MCDB	16.2	16.7	18.9	22.8	27.8	31.6	31.8	32.0	28.0	24.9	21.3	18.2
	5%	WB	12.4	12.6	13.5	15.2	18.2	20.9	22.0	22.1	20.9	19.5	16.9	14.1
		MCDB	15.3	15.9	17.8	21.5	26.4	31.0	31.9	31.6	27.8	24.1	20.4	16.7
	10%	WB	11.4	11.6	12.6	14.4	17.5	20.3	21.4	21.4	20.2	18.5	15.8	12.9
		MCDB	14.3	14.9	16.9	20.5	25.2	30.1	31.7	31.3	27.6	23.2	19.4	15.7
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	8.9	9.9	10.5	10.6	10.2	9.7	9.3	8.2	7.7	7.3
		MCDBR	9.5	10.2	11.5	12.5	12.3	12.4	12.0	11.3	11.1	10.1	9.5	9.3
		MCWBR	6.1	6.3	6.1	6.0	5.2	4.7	4.8	4.7	4.9	4.9	5.5	6.1
	5% WB	MCDBR	8.4	9.2	10.1	11.4	11.3	11.1	10.9	10.4	10.3	9.1	8.4	8.1
		MCWBR	5.9	6.2	6.1	6.0	5.4	4.9	4.9	4.9	5.0	5.1	5.3	5.8
Clear-Sky Solar Irradiance	taub	0.355	0.380	0.418	0.463	0.445	0.425	0.429	0.436	0.432	0.404	0.382	0.365	
	taud	2.408	2.317	2.207	2.092	2.207	2.303	2.301	2.272	2.278	2.332	2.357	2.378	
	Ebn at Noon	814	838	838	820	842	857	850	836	815	800	772	770	
	Edh at Noon	88	108	133	158	143	130	129	130	122	105	92	85	
All-Sky Solar Radiation	RadAvg	1.96	2.74	4.09	5.42	6.49	7.43	7.57	6.74	5.09	3.44	2.30	1.70	
	RadStd	0.23	0.34	0.34	0.48	0.33	0.39	0.18	0.16	0.29	0.30	0.24	0.24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+0.50	N/A	N/A	+0.55	N/A	N/A	N/A	-85	N/A	N/A	
Regional (12 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-72	+85	N/A	

Nomenclature: See separate page